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Topology-aware unsupervised domain adaptation for curvilinear structure segmentation

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ABSTRACT

Unsupervised domain adaptation (UDA) enables segmentation models to generalize to unlabeled clinical domains. However, existing methods primarily address appearance-level discrepancies, while neglecting topological mismatches in connectivity, branching, and loop structure-characteristics of curvilinear anatomies. This oversight frequently leads to fragmented, clinically unreliable segmentations of structures such as vascular networks. To bridge this gap, we propose TopoUDA, a framework that explicitly integrates topological reasoning at multiple stages of adaptation. First, we design a Curvilinear-Aware Discriminator (CAD), an adversarial critic that distinguishes anatomically plausible structures from fragmented predictions based on a compact topological fingerprint. To enable end-to-end training, we introduce a differentiable proxy network that provides a valid gradient path through this otherwise non-differentiable structural assessment. Second, we develop a Hierarchical Topological Refiner (HTR) that leverages persistent homology to identify stable structural components and refine spatially related fragments, generating structurally coherent pseudo-labels. Third, we introduce a Hybrid Trust Mechanism (HTM) that combines appearance- and topology-based reliability scores from dual discriminators to adaptively weight pseudo-label supervision, ensuring the model learns primarily from anatomically plausible predictions. Extensive validation across retinal fundus photography, Optical Coherence Tomography Angiography (OCTA), and electron microscopy demonstrates that TopoUDA significantly outperforms state-of-the-art methods in both overlap and topological metrics. Furthermore, qualitative generalization on the large-scale, unlabeled AI-READI cohort underscores TopoUDA's practical applicability to real-world clinical data where ground-truth annotations are unavailable.

1. Introduction

Segmentation of curvilinear structures, such as vascular networks, neuronal processes, and airway trees, is a fundamental task in medical image analysis, underpinning quantitative assessment in a wide range of clinical and scientific applications like ophthalmology, neuroimaging, and pulmonology (Zhao et al., 2023; Zhuang et al., 2024). These structures form intricate, spatially connected filaments where morphology (e.g., tortuosity, branching patterns) directly encodes critical biomarkers for diseases like diabetic retinopathy and Retinopathy of Prematurity (ROP) (Fraz et al., 2012a). However, manual annotation of these complex networks is prohibitively time-consuming and prone to variability, especially in high-resolution modalities like Optical Coherence Tomography Angiography (OCTA) and Electron Microscopy (EM). While Convolutional Neural Networks (CNNs) have achieved state-of-the-art performance when abundant labels are available (Ronneberger et al., 2015; Zhuang et al., 2024; Lei et al., 2021; Moccia et al., 2018; Khadangi et al., 2021; Fakhry et al., 2016), they remain notoriously brittle when applied outside their training domain.

This performance degradation is driven by domain shift, which arises from two distinct sources. The first is the *Appearance Gap*, caused by variations in imaging devices, noise characteristics, and patient populations (Wilson and Cook, 2020; Wu et al., 2023). The second, more subtle but often more damaging, is the *Structural Gap*, inherent differences in the geometric and topological organization of the anatomy of curvilinear networks itself. For instance, retinal vessels in preterm infants are significantly more tortuous and underdeveloped than those in adults. While the Appearance Gap degrades local boundary precision,

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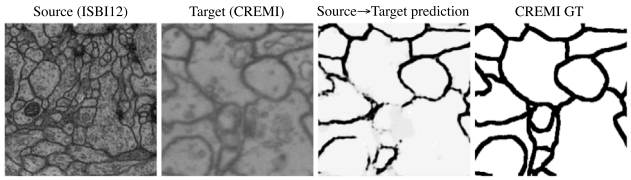


Fig. 1. Domain shift and structural degradation in electron microscopy segmentation. Training on ISBI12 and testing directly on CREMI leads to severe structural errors: despite locally confident predictions, the source-only model produces fragmented membrane boundaries and broken contours. This highlights the challenge of curvilinear adaptation and the need for structure-aware UDA.

the Structural Gap leads to globally inconsistent predictions, including broken branches and disconnected components, that violate the network's intrinsic connectivity. This failure is illustrated in Fig. 1, where a model trained on one topology produces locally confident yet globally fragmented predictions on a structurally shifted target.

Unsupervised Domain Adaptation (UDA) offers a principled framework to bridge these gaps in domain shift by transferring knowledge from a labeled source domain to an unlabeled target domain (Ganin et al., 2016; Tzeng et al., 2017; Zhang et al., 2019). Dominant strategies, such as adversarial alignment (Vu et al., 2019) and pseudo-labeling (Zou et al., 2018), have proven effective for compact objects (e.g., organs, tumors) where aligning visual appearance largely suffices. However, for curvilinear structures, these standard approaches fall short. Their methods lack mechanisms to explicitly preserve the structural constraints. As demonstrated in Fig. 2, even state-of-the-art UDA methods like ADVENT (Vu et al., 2019), FDA (Yang and Soatto, 2020), SFDA-DPL (Wu et al., 2023), and CADA (Ma et al., 2023) fail to preserve connectivity. This distinction is crucial: while Fig. 1 shows that *no adaptation* leads to failure, Fig. 2 reveals that *standard adaptation* also fails because it only aligns texture (Appearance Gap) without enforcing topological constraints (Structural Gap).

We focus our investigation on two primary UDA paradigms, adversarial learning (Haq, 2020; Tzeng et al., 2017; Vu et al., 2019) and pseudo-labeling (Choi et al., 2019; Wu et al., 2023; Li et al., 2023), as they represent the effective levers for adaptation, yet currently suffer from a shared limitation: pixel-centricity. Standard adversarial discriminators evaluate visual realism rather than structural coherence, allowing anatomically implausible predictions to pass undetected. Similarly, pseudo-labeling strategies rely on pixel-level confidence (Zou et al., 2018; Choi et al., 2019; Zheng and Yang, 2021), which neglects global topology; a locally confident pixel can still belong to a globally inconsistent branch. To bridge the Structural Gap, it is therefore necessary to re-imagine these paradigms: transforming adversarial alignment to distinctively evaluate connectivity, and upgrading pseudo-labeling to filter candidates based on topological stability rather than just pixel intensity.

To achieve this, we propose TopoUDA, a topology-aware unsupervised domain adaptation framework designed to explicitly address both the Appearance Gap and the Structural Gap. The framework integrates three synergistic components. First, a dual-critic adversarial system aligns predictions in both pixel and composite structural feature spaces, capturing 16 distinct topology and geometry metrics. Second, a Hierarchical Topological Refiner (HTR) generates structurally consistent pseudo-labels guided by persistent homology, identifying stable structures (Level 1) and rescuing spatially related contour fragments (Level 2). Finally, a Hybrid Trust Mechanism (HTM) re-weights supervision using both appearance- and structure-based confidence. This unified design enables structure-aware adaptation across diverse curvilinear modalities. Our main contributions are:

1. We introduce *TopoUDA*, a topology-aware unsupervised domain adaptation framework for curvilinear structure segmentation, explicitly distinguishing between the Appearance Gap and the Structural Gap, and demonstrating that addressing appearance alone is insufficient for vascular and neuronal networks.
2. We propose a structure-guided adaptation strategy that integrates global topological priors into both domain alignment and pseudo-label learning, enabling the preservation of connectivity, continuity, and branching patterns during cross-domain transfer without requiring target-domain annotations.
3. We demonstrate the effectiveness of TopoUDA across nine UDA tasks in three medical imaging modalities: retinal fundus, OCTA, and electron microscopy. Beyond achieving superior overlap and structural integrity metrics, its performance is further validated qualitatively on a large-scale unlabeled OCTA dataset.

2. Related work

2.1. Domain adaptation for medical image segmentation

UDA for semantic segmentation seeks to reduce the generalization gap that arises when a segmentation model trained on labeled source data fails to perform on unlabeled target data. The prevailing paradigm is adversarial alignment, in which a discriminator distinguishes between source and target representations while the encoder learns to make them indistinguishable (Ganin et al., 2016). Later works extended this principle to the output space, aligning predicted segmentation maps rather than latent features (Tsai et al., 2018). In parallel, distance-based approaches such as Maximum Mean Discrepancy (MMD) and Correlation Alignment (CORAL) (Gretton et al., 2012; Sun and Saenko, 2016) perform statistical distribution matching without adversarial optimization.

These methods primarily address appearance-level discrepancies such as contrast, color, or texture, which we refer to as the Appearance Gap. Yet in medical imaging, the Structural Gap often dominates: differences in anatomical layout, connectivity, or morphology that cannot be resolved by texture alignment alone. Standard adversarial or statistical methods are structurally agnostic; they do not preserve geometric or topological consistency. Some recent studies introduced boundary-level regularization (Zhang et al., 2024), but such local constraints fail to capture the global topology of filamentary structures like vessels or neurons, where continuity and branching integrity define the anatomy. These observations suggest that domain adaptation for curvilinear anatomy requires explicit modeling of structural and connectivity priors, beyond appearance-based alignment alone.

More recent extensions such as Source-Free Domain Adaptation (SFDA) (Kundu et al., 2022; Li et al., 2025) and Test-Time Adaptation (TTA) (Wang et al., 2021) remove access to source data entirely, heightening the need for strong structural priors. Within this line, AdaptDiff (Hul et al., 2024) leverages diffusion-based image translation to synthesize target-like training data, and AIF-SFDA (Li et al., 2025) performs amplitude-and-frequency disentanglement to align source-free representations. Even recent topology-guided TTA methods such as TopoTTA (Huang et al., 2025) demonstrate the utility of topological priors but remain limited to single-image, online adaptation rather than full-dataset UDA. Hence, effectively adapting across domains while preserving fine-grained topological coherence remains an open problem. Our work directly addresses this gap by introducing explicit topological reasoning into the UDA process.

2.2. Pseudo-labeling in UDA

Self-training constitutes the second major UDA paradigm, where the model generates pseudo-labels on the target domain and retrains on them. The success of this process critically depends on pseudo-label quality.

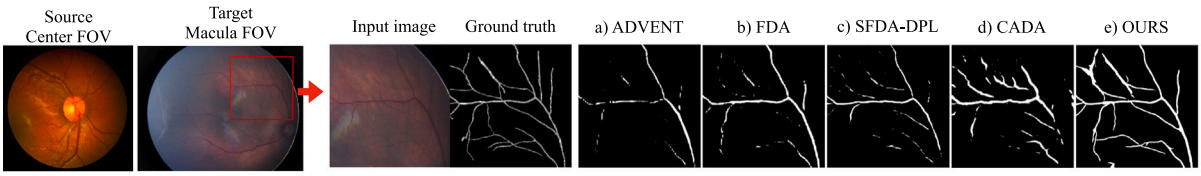


Fig. 2. Motivating examples for topology-aware UDA. Segmentation results on a challenging ROP target image with a macula Field of View (FOV), a significant structural shift from the optic disc-centered FOV and adolescent source domain. Structurally agnostic baselines like (c) FDA and (d) SFDA-DPL produce topologically flawed results, failing to preserve the vessel network's connectivity. In contrast, (e) our TopoUDA, by explicitly modeling topological priors, successfully eliminates the spurious component and bridges the structural gap, demonstrating its ability to solve the "Structural Gap".

A simple strategy is to threshold high-confidence pixels in a Mean-Teacher (Tarvainen and Valpola, 2017) or consistency framework, which often yields sparse, incomplete supervision, especially for thin or low-contrast structures.

Recent works refine pseudo-label quality using consistency regularization (Mittal et al., 2019), uncertainty-based weighting via Monte Carlo Dropout or ensembles (Wu et al., 2023; Zheng and Yang, 2021), or multi-model consensus and progressive densification (Pan et al., 2024; Chen et al., 2023). While these techniques suppress pixel-level noise, they remain local in nature evaluating reliability per pixel or patch rather than assessing global structural plausibility. Consequently, pseudo-labels may appear confident yet violate anatomical continuity, reinforcing false connections or broken branches. Our approach explicitly overcomes this limitation by refining pseudo-labels according to topological significance and hierarchical vessel organization.

2.3. Topological Data Analysis (TDA) in medical imaging

TDA offers a mathematical lens for quantifying structure in images beyond pixel intensity. Persistent Homology (PH) summarizes topological features, connected components (H_0) and loops (H_1), across scales, providing robustness to noise and deformation (Motlagh et al., 2020). In segmentation, TDA has informed shape-aware loss functions that penalize topological errors (Hu et al., 2019; Shit et al., 2021; Yang et al., 2021b,a), and differentiable formulations now enable end-to-end training using discrete Morse theory or differentiable persistence (Mosinska et al., 2021).

Within the *supervised* setting, several lines of work have explored topology-aware segmentation. clDice (Shit et al., 2021) introduces a centerline-Dice loss that preserves connectivity at the single-image level. TR-GAN (Chen et al., 2020) employs a topology-ranking discriminator for artery/vein classification under full supervision. Hu et al. (2019) integrate Betti-number constraints into the training loss using persistent homology. DTU-Net (Lin et al., 2023) explicitly disentangles appearance and topology branches during supervised training, and Mosinska et al. (2018) use pretrained VGG-16 features as a perceptual surrogate for topological correctness. These works establish that explicit topological signals can improve supervised segmentation, but they assume access to target-domain labels. TopoUDA carries this insight into the unsupervised setting, where topology must be enforced without ground-truth supervision on the target domain.

The integration of TDA into adaptation settings is only beginning to emerge. The recent TopoTTA framework (Huang et al., 2025) employs a topological loss for per-image Test-Time Adaptation, showing that topology can guide self-supervised refinement. However, its scope is limited to single-sample correction. In contrast, we extend this idea to full-dataset Unsupervised Domain Adaptation, where topological reasoning informs both pseudo-label generation and adversarial alignment. By leveraging persistent homology within a global adaptation loop, TopoUDA enforces structural coherence across the entire target domain.

3. Method

We consider the UDA setting for curvilinear structure segmentation, where the goal is to transfer segmentation knowledge from a labeled source domain to an unlabeled target domain. The labeled source domain is denoted as $D_S = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, where $x_i^s \in \mathbb{R}^{HxWxC}$ and $y_i^s \in \{0,1\}^{HxW}$. The unlabeled target domain $D_T = \{(x_i^t)\}_{i=1}^{N_t}$ contains only images $x_i^t \in \mathbb{R}^{HxWxC}$, reflecting the common clinical setting where manual annotation is unavailable or prohibitively expensive. Since source and target domains are drawn from different distributions, the challenge is to learn a segmentation network S that generalizes to D_T by exploiting labeled supervision from D_S and structural cues from unlabeled target data.

As illustrated in Fig. 3, TopoUDA addresses this challenge through four integrated components: (1) source-supervised warm-up with feature regularization (Section 3.1), (2) a dual-critic adversarial module enforcing both appearance and structural alignment (Section 3.2), (3) a Hierarchical Topological Refiner generating structure-consistent pseudo-labels (Section 3.3), and (4) a Hybrid Trust Mechanism for adaptive pseudo-label weighting (Section 3.4). Section 3.5 presents the unified learning objective.

3.1. Source warm-up and feature regularization

We first train the segmentation network S on the labeled source dataset D_S to predict curvilinear probability maps $\hat{y}^s = S(x^s)$. The supervised objective combines Binary Cross-Entropy (BCE) and Dice losses:

$$\mathcal{L}_{\text{sup}} = \text{BCE}(\hat{y}_i^s, y_i^s) + \text{Dice}(\hat{y}_i^s, y_i^s), \tag{1}$$

To encourage early domain-invariant representations and improve stability during subsequent adaptation, we additionally apply a lightweight feature regularization term based on Maximum Mean Discrepancy (MMD) (Gretton et al., 2012). We sample paired mini-batches of labeled source images x^s and unlabeled target images x^t, and compute:

$$\mathcal{L}_{\text{feat}} = \text{MMD}\big(f(S(x_i^s)), f(S(x_i^t))\big), \tag{2}$$

where $f(\cdot)$ denotes L_2-normalized bottleneck features. The overall warm-up objective is:

$$\mathcal{L}_{\text{warm}} = \mathcal{L}_{\text{sup}} + \lambda_{\text{feat}}\mathcal{L}_{\text{feat}}. \tag{3}$$

This warm-up phase initializes S with strong source supervision while gently aligning source and target feature distributions, reducing instability when unlabeled target supervision is introduced.

3.2. Dual-critic adversarial module

TopoUDA employs two complementary discriminators: a pixel-level discriminator D_{pixel} enforcing appearance alignment, and a Curvilinear-Aware Discriminator D_{topo} enforcing structural consistency.

Pixel-Level Discriminator - D_{pixel}**.** D_{pixel} operates on segmentation probability maps and discriminates between source and target outputs

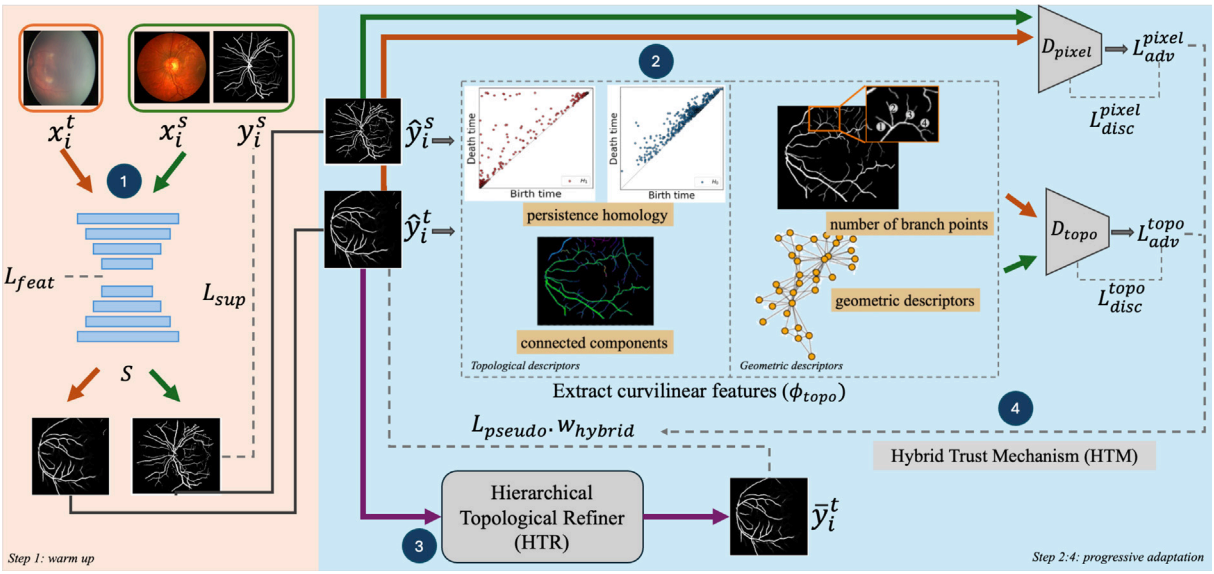


Fig. 3. Overview of the proposed TopoUDA framework. The system integrates three core modules: a segmentation network S, dual-critic discriminators $\{D_{pixel}, D_{topo}\}$, and topology-guided pseudo-label refinement. Training proceeds iteratively through four steps: **(1)** Source supervising and feature regularization warm-up, where S is optimized with supervised learning and feature alignment to promote domain-invariant representations; **(2)** Dual-critic adversarial alignment, in which a pixel-level discriminator enforces appearance realism and a Curvilinear-Aware Discriminator (CAD) enforces structural consistency; **(3)** Hierarchical Topological Refinement (HTR), which applies persistent homology to extract stable anchors and optionally rescue spatially coherent contour fragments using ϕ_{topo}; and **(4)** Hybrid Trust Mechanism (HTM), which computes appearance- and structure-driven trust scores that contribute to the pseudo-label loss L_{pseudo} Iterative execution progressively reduces both the Appearance Gap and the Structural Gap between source and target domains.

based on low-level appearance statistics such as texture and local shape. It adopts a standard CNN architecture with three blocks: convolution, batch normalization, and activation.

Given source and target images $x^s \sim D_S$ and $x^t \sim D_T$, the discriminator loss is defined as:

$$\mathcal{L}_{disc}^{pixel} = -\mathbb{E}_{x^s \sim D_S}[\log D_{pixel}(S(x^s))] - \mathbb{E}_{x^t \sim D_T}[\log(1 - D_{pixel}(S(x^t)))]. \quad (4)$$

The segmentation network S is updated using the corresponding adversarial objective on target-domain predictions:

$$\mathcal{L}_{adv}^{pixel} = \mathbb{E}_{x^t \sim D_T}[-\log D_{pixel}(S(x^t))], \quad (5)$$

which encourages target predictions to exhibit appearance statistics similar to source predictions in the discriminator's feature space.

Curvilinear-Aware Discriminator - D_{topo}**.** D_{topo} evaluates structural plausibility using compact *topological and geometric descriptors* that capture connectivity, continuity, and branching patterns.

However, since topological and many geometric measurements are inherently non-differentiable, directly back-propagating structural feedback to the segmentation network is not feasible. To enable end-to-end training, we introduce a differentiable proxy (ProxyNet) that approximates these descriptors, allowing structural gradients from D_{topo} to guide the segmentation network during adaptation.

(1) Topo-Geometric Feature Representation:

We define a 16-dimensional feature vector $\phi_{topo}(\hat{y})$. These descriptors jointly capture global connectivity patterns and local morphological characteristics, forming a compact structural fingerprint for curvilinear networks. The feature vector includes:

Topological descriptors: The Betti numbers (β_0, β_1) computed from a 2D cubical complex on the predicted mask $\hat{y}$ (connected components and loops), the total H_1 persistence (sum of loop persistences), and the entropy of the persistence distribution.

Geometric/skeleton descriptors: total skeleton length, number of branch points, junction density, a 7-bin degree-distribution histogram of skeleton node valences, mean branch curvature, and caliber variation along the vessels.

In total, this yields 4 topological features (Betti numbers β_0, β_1, total H_1 persistence, persistence entropy) and 12 geometric/skeleton features, resulting in a 16-dimensional descriptor. All 16 features are used as input to the curvilinear-aware discriminator D_{topo}. To improve numerical stability and ensure comparable scaling across descriptors, we standardize ϕ_{topo} using statistics computed on the source domain. Specifically, we estimate a mean vector $\boldsymbol{\mu} \in \mathbb{R}^{16}$ and standard deviation vector $\boldsymbol{\sigma} \in \mathbb{R}^{16}$ over the source training set, and apply feature-wise standardization. The resulting standardized feature vector $\tilde{\phi}_{\text{topo}}(\hat{y})$ is then fed into D_{topo}.

(2) Proxy Network:

To bridge the gap between non-differentiable structural measurements and gradient-based learning, we introduce a lightweight proxy network (ProxyNet) $P_\psi : [0,1]^{H \times W} \to \mathbb{R}^{16}$ which learns to approximate the mapping from a predicted segmentation mask to its corresponding topo-geometric feature vector.ProxyNet is implemented as a compact CNN with four strided-convolution blocks (channels 16, 32, 64, 128; ReLU activations) followed by global average pooling and two fully-connected layers ($128 \to 64 \to 16$), totaling 106,574 parameters ($\approx$0.107 M).

ProxyNet is pre-trained on the labeled source domain using a regression objective that matches the true structural descriptors computed from ground-truth masks.

$$\mathcal{L}_{proxy} = \mathbb{E}_{(x^s, y^s) \sim D_S}\left[\|P_\psi(S(x^s)) - \phi_{\text{topo}}(y^s)\|_2^2\right]. \quad (6)$$

Once trained, P_ψ is frozen and used during adaptation as a differentiable surrogate for ϕ_{topo}. This enables gradients from D_{topo} to propagate back to the segmentation network in an end-to-end fashion during the adaptation phase.

(3) Adversarial Learning:

Finally, we establish an adversarial game where D_{topo} is trained to distinguish the topo-geometric fingerprints of ground truth masks from those generated by the segmentation network S. Let $\phi_{topo}(y^s)$ denote the real feature vectors computed from source-domain annotations, and $P_\psi(S(x^t))$ represents differentiable predicted feature vectors from ProxyNet. The discriminator is optimized via the adversarial objective:

$$\begin{aligned}\mathcal{L}_{disc}^{topo} = &-\mathbb{E}_{y^s \sim D_S}\left[\log D_{\text{topo}}(\phi_{\text{topo}}(y^s))\right] \\ &-\mathbb{E}_{x^t \sim D_T}\left[\log(1 - D_{\text{topo}}(P_\psi(S(x^t))))\right].\end{aligned} \quad (7)$$

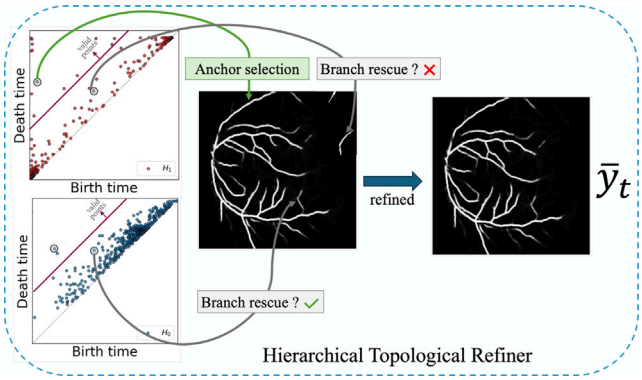


Fig. 4. Persistent homology identifies high-persistence components as anchors, while lower-persistence components are evaluated for spatial proximity and rescued when forming valid contour fragments. The final pseudo-label merges anchors and rescued branches, filtering topological noise while preserving fine curvilinear structure.

The segmentation network S is trained adversarially to produce curvilinear masks whose structural fingerprints can fool D_{topo} thereby aligning structural characteristics across domains.

The segmentation network's adversarial loss for D_{topo} is:

$$L_{adv}^{topo} = \mathbb{E}_{x^t \sim D_T}[-\log D_{topo}(P_\psi(S(x^t)))]. \tag{8}$$

By combining the pixel-level and curvilinear-aware discriminators, TopoUDA jointly aligns appearance-level and structure-level distributions. This dual-critic mechanism enforces both visual realism and curvilinear coherence in an end-to-end differentiable framework, effectively bridging the appearance and structural gaps between domains.

3.3. Hierarchical Topological Refiner (HTR)

Conventional pseudo-labeling strategies rely primarily on pixel-level confidence, disregarding whether predicted structures remain globally coherent. As a result, locally confident pixels may belong to broken, merged, or anatomically implausible branches. Our HTR mitigates this limitation by explicitly incorporating structural reasoning into pseudo-label generation through persistent-homology analysis and spatial hierarchy modeling.

Persistent Homology Analysis. Persistent homology (PH) (Edelsbrunner and Harer, 2010) tracks the evolution of topological features across a filtration of an image. It identifies two primary feature classes: connected components (H_0), representing emerging branches, and loops (H_1), corresponding to enclosed ring-like structures. Each feature is represented as a birth–death pair (b, d) in a persistence diagram, with persistence $|d - b|$ quantifying structural stability. High-persistence features correspond to robust, large-scale structures (e.g., major vessels), while low-persistence features often represent noise.

HTR applies this framework directly to the predicted probability map $\hat{y} = S(x^t)$ to probe its underlying topology across multiple scales. We construct a cubical complex by thresholding $\hat{y}$ at discrete levels $\alpha \in [0, 1]$ to form a nested sequence of binary masks:

$$B_\alpha = \{(i, j) : \hat{y}(i, j) \geq \alpha\}, \quad B_{\alpha_1} \supseteq B_{\alpha_2} \supseteq \cdots \supseteq B_{\alpha_K}.$$

For each extracted component c, we compute its persistence and normalize it for scale invariance:

$$\text{norm_persistence}(c) = \frac{persistence(c)}{\max(persistence) - \min(persistence)}. \tag{9}$$

Crucially, each persistence point is spatially traceable to a specific connected component in the image domain. During filtration, we maintain a lookup table that links each topological feature to its corresponding pixel coordinates, enabling spatial reasoning between persistent structures. Fig. 4 provides an intuitive visualization of this process, linking persistence values in the diagram to their spatial realizations, a property that enables HTR to preserve fine structural details while suppressing topological noise.

Two-Level Anchor-and-Rescue Mechanism. HTR emulates the hierarchical nature of curvilinear networks where prominent trunks or backbones support finer branches through a two-level refinement process.

Level 1 - Stable-Structure Extraction: High-persistence components are selected as structural anchors. The anchor set includes components satisfying either H_0 or H_1 persistence criteria:

$$c_{\text{anchor}} = \{\ c \in C\ :\ \text{norm_persistence}(c) \geq \tau_{H_0} \text{ or } \text{norm_persistence}(c) \geq \tau_{H_1}\ \}, \tag{10}$$

where C is the set of all detected components and τ_{H_0}, τ_{H_1} denote persistence thresholds for connected components and loops, respectively. These anchors correspond to the most stable and structurally dominant elements such as major curvilinear trunks or closed rings.

Level 2 - Context-Aware Refinement: We then consider moderate persistence candidates that fail to meet the anchor criteria:

$$c_{cond} = \{c \in C \setminus C_{anchor} : \text{norm_persistence}(x) \geq \tau_{rescue}\}, \tag{11}$$

where $\tau_{rescue} < min(\tau_{H_0}, \tau_{H_1})$ is a lower rescue threshold. A candidate component c is rescued if it lies spatially close to any anchor component:

$$c_{rescued} = \{c \in C_{cond} : \min_{a \in C_{anchor}} d_{spatial}(c, a) \leq \delta_{rescue}\}. \tag{12}$$

The spatial distance between components is computed as:

$$d_{spatial}(c_1, c_2) = \min_{p_1 \in c_1, p_2 \in c_2} \|p_1 - p_2\|_2. \tag{13}$$

This hierarchical selection ensures that faint but anatomically valid branches connected to dominant structures are preserved, while isolated or noisy fragments are suppressed.

Refined Pseudo-Label Generation. HTR operates on the probability map $\hat{y}^t$ to identify persistent structures, while the output pseudo-label $\bar{y}^t$ is a binary mask. The final pseudo-label integrates both stable anchors and rescued components:

$$\bar{y}^t = (c \in C_{anchor}) \cup (c \in C_{rescued}). \tag{14}$$

This process ensures that $\bar{y}^t$ preserves major structural elements while maintaining connectivity through validated minor branches, effectively filtering topological noise while preserving fine structural details essential for curvilinear networks. The HTR algorithm generates pseudo-labels that are both locally confident and globally consistent, addressing the fundamental limitation of pixel-confidence-based approaches in curvilinear structure segmentation.

3.4. Hybrid Trust Mechanism (HTM)

While our HTR generates structurally coherent pseudo-label candidates, a critical challenge remains: ensuring their trustworthiness for model training. Conventional UDA frameworks often rely solely on pixel-level confidence or hard thresholding to assess pseudo-label quality, which may select predictions that appear locally plausible yet violate structural constraints. Our Hybrid Trust Mechanism (HTM) addresses this by combining assessments from both discriminators to compute adaptive quality scores that ensure pseudo-labels satisfy both appearance and structural criteria.

Trust Score Computation. For a target image x^t with generator prediction $\hat{y}^t = S(x^t)$ and refined pseudo-label $\bar{y}^t$ from HTR, we compute two complementary trust scores that capture different aspects of prediction quality.

The *appearance trust* measures how realistic the prediction appears in terms of visual characteristics:

$$w_{pixel} = \sigma(D_{pixel}(\hat{y}^t)). \tag{15}$$

The *structural trust* evaluates the topological plausibility of the prediction using our differentiable proxy:

$$w_{topo} = \sigma(D_{topo}(P_{\psi}(\hat{y}^t))), \tag{16}$$

where $\sigma(.)$ is the sigmoid function ensuring both scores are in [0,1].

Hybrid Trust Score. We combine both assessments using element-wise multiplication, implementing a conservative trust strategy where pseudo-labels must satisfy both criteria to receive high confidence:

$$w_{hybrid} = w_{pixel} \times w_{topo}, \tag{17}$$

where w_{hybrid} is a scalar trust weight per image. This formulation ensures that only predictions passing both appearance and structural validation contribute significantly to model training. When $w_{pixel} \approx 1$ and $w_{topo} \approx 1$, the pseudo-label receives full supervision weight. Conversely, if either discriminator indicates low quality, the hybrid score approaches zero, minimizing the contribution of potentially harmful training signals.

The trust-weighted pseudo-label loss becomes:

$$\mathcal{L}_{pseudo} = \mathbb{E}_{x^t \sim D_T}\left[w_{hybrid} \cdot L_{sup}(\hat{y}^t, \tilde{y}^t)\right], \tag{18}$$

where L_{sup} combines Binary Cross-Entropy and Dice loss as defined in Section 3.1. This soft weighting approach prevents the complete rejection of potentially useful training signals while prioritizing structural and visual consistency, enabling robust adaptation even when perfect pseudo-labels are unavailable.

3.5. Learning objective

To jointly address the Appearance and Structural Gaps while maintaining source-domain performance, TopoUDA minimizes a unified objective function that integrates supervision, regularization, and dual-adversarial alignment. The total loss $\mathcal{L}_{\text{total}}$ is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{warm}} + \lambda_{\text{adv}}(\mathcal{L}_{\text{adv}}^{\text{pixel}} + \mathcal{L}_{\text{adv}}^{\text{topo}}) + \lambda_{\text{pseudo}}\mathcal{L}_{\text{pseudo}}, \tag{19}$$

where $\mathcal{L}_{\text{warm}}$ (Eq. (3)) anchors the model to the source domain via supervised and MMD regularization, preventing catastrophic forgetting. The adversarial terms $\mathcal{L}_{\text{adv}}^{\text{pixel}}$ (Eq. (5)) and $\mathcal{L}_{\text{adv}}^{\text{topo}}$ (Eq. (8)) drive the network to generate target predictions that are indistinguishable from source labels in both texture and topology. Finally, $\mathcal{L}_{\text{pseudo}}$ (Eq. (18)) leverages the HTR-refined masks to propagate structural knowledge back into the network, weighted by our hybrid trust scores.

Training proceeds in an alternating optimization framework. In each iteration, we first update the discriminators D_{pixel} and D_{topo} to distinguish source from target predictions. Subsequently, the segmentation network S is updated to minimize $\mathcal{L}_{\text{total}}$, effectively fooling the dual critics while satisfying the structural constraints imposed by the high-confidence topological pseudo-labels. This multi-objective optimization ensures that the model progressively aligns domains across pixel, feature, and structural levels simultaneously.

4. Experiments

To evaluate the effectiveness of the proposed framework across diverse medical curvilinear domains, we conducted experiments on nine source–target domain pairs spanning three modalities: retinal fundus photography, OCTA, and EM. Each modality exhibits unique domain shifts, ranging from appearance variations caused by different acquisition devices and patient populations in retinal imaging, to contrast and noise differences across OCTA scanners, and staining or resolution variations in EM datasets. These settings were carefully selected to emulate real-world deployment conditions where the source and target share anatomical similarity but differ in imaging characteristics, as illustrated in Fig. 1.

4.1. Datasets

Retinal Fundus. We utilize the public datasets DRIVE (Staal et al., 2004), STARE (Hoover et al., 2000), and CHASEDB1 (Fraz et al., 2012b), along with an in-house dataset of preterm ROP fundus images. These datasets differ in both image acquisition and population demographics: DRIVE and STARE include adult subjects with standard field-of-view and color calibration, while CHASEDB1 images include adolescent subjects, and ROP images capture preterm neonates with narrower field-of-view and immature vasculature. The cross-domain experiments were performed on five representative pairs: STARE → CHASE, DRIVE → CHASE, DRIVE → STARE, CHASE → STARE, and CHASE→ ROP.

Optical Coherence Tomography Angiography (OCTA). To assess generalization under modality-specific noise and resolution differences, we used OCTA-500 (Li et al., 2024) and ROSE-1 (Ma et al., 2020), two public datasets acquired using different scanners and imaging protocols. Both datasets contain microvascular en face angiograms with manually annotated vessel masks. We conducted bidirectional adaptation (ROSE → OCTA-500, OCTA-500 → ROSE) to analyze the robustness of TopoUDA under cross-device appearance variation.

In addition, we evaluated TopoUDA qualitatively on a large-scale clinical OCTA dataset from AI-READI project for the study of type 2 diabetes. This dataset contains diverse real-world scans but lacks pixel-level annotations for training or direct quantitative comparison. This experiment serves as an external qualitative validation, illustrating the framework's ability to produce structurally coherent segmentations in realistic, unlabeled clinical settings. Beyond validation, this work offers a practical contribution to the AI-READI project by generating high-quality initial annotations for this unlabeled cohort, potentially facilitating in-depth studies of OCTA biomarkers to advance fundamental understanding of microvascular injury and remodeling across retinal vascular diseases.

Electron Microscopy (EM). We further validated our method on high-resolution neuron segmentation datasets, CREMI (Funke et al., 2016) and ISBI12 (Arganda-Carreras et al., 2012), representing serial-section EM of neural tissue. These datasets differ in sample preparation, staining thickness, and contrast, offering a rigorous test for structural consistency across severe domain gaps. The adaptation was evaluated bidirectionally (ISBI12→ CREMI and CREMI→ ISBI12).

Structural variation across domains. Beyond appearance differences, the datasets above exhibit substantial *structural* variation: connectivity patterns (β_0, β_1), skeleton lengths, branch-point counts, and junction densities differ considerably across domains (Table 1). The most pronounced shifts are: (i) the retinal fundus triad (STARE/CHASE / ROP), where STARE has the densest junction structure (junction density 54.5, branch points 476), CHASE is intermediate (26.0, 298), and ROP is the sparsest (14.3, 104), reflecting the less-arborized vasculature of preterm neonates; only ROP exhibits non-zero mean β_1 (1.8 loops per image); (ii) ROSE-1 vs. OCTA-500, where ROSE-1 shows 5.3× more branch points and 3.1× higher junction density, consistent with denser micro-capillary annotation; and (iii) CREMI vs. ISBI12, which differ substantially in β_1 and in branch-point density due to differences in annotation protocol. The large discrepancy in β_0 between ISBI12 (1681.7) and CREMI (5.8) reflects that ISBI12 uses thinner membrane lines that fragment into many small connected components under 8-connectivity, while CREMI's thicker, smoother membrane annotations form a single connected tessellation; this annotation-protocol difference is itself a form of domain shift that the segmentation network must bridge alongside modality-level appearance differences. This structural heterogeneity motivates the explicit topology-aware adaptation strategy adopted by TopoUDA.

Table 1
Topological and geometric characteristics of source and target domains, computed on the ground-truth masks of each test set. β_0 is the mean number of 8-connected foreground components per image; β_1 is the mean number of independent loops per image (computed via the Euler characteristic on the cubical foreground complex); "Skel. Len." is mean skeleton length (pixels); "Branch Pts." is mean number of skeleton branch points; "Junc. Dens." is junction density (branch points per 1000 skeleton pixels). Datasets are grouped by modality.

Dataset	β_0	β_1	Skel. Len.	Branch Pts.	Junc. Dens.
STARE	6.0	0.0	8466	476	54.5
CHASEDB	2.7	0.0	11,389	298	26.0
ROP	4.5	1.8	7045	104	14.3
OCTA-500	9.1	0.0	5226	310	58.5
ROSE-1	17.8	0.0	8291	1629	183.8
ISBI12	1681.7	661.7	39,811	18,039	451.8
CREMI	5.8	282.7	37,358	2162	57.9

4.2. Preprocessing

All datasets were preprocessed, following a unified protocol. Images were normalized to the [−1,1] intensity range and patched into 400 × 400 pixel patches for training. Standard data augmentation techniques including random flips, 90° rotations, and intensity jittering, were applied to improve generalization.

Following standard UDA practice (Zou et al., 2018; Vu et al., 2019; Hoyer et al., 2022), all methods are trained once per source→target domain pair without cross-validation, since target labels are unavailable and cross-validation on unlabeled data is not meaningful in this setting. For datasets that provide predefined training/testing splits (e.g., DRIVE, CHASEDB, STARE, ISBI12), we strictly follow their official partitions for all methods. For datasets without predefined splits (e.g., CREMI, OCTA-500), we adopt a fixed 85/15 train–test split, and this exact split is shared across all baselines and TopoUDA to ensure consistent comparison. This strategy reflects standard UDA evaluation protocols and maintains full experimental reproducibility. All hyperparameters and training schedules were tuned on a held-out subset of the source domain based on segmentation performance and then fixed across all source→target pairs and modalities.

4.3. Evaluation metrics

To comprehensively evaluate adaptation performance, we employed both region-based and structure-based segmentation metrics.

Region-based Performance. We measured standard overlap metrics including the Dice coefficient and Intersection-over-Union (IoU) to quantify overall segmentation agreement between predictions and ground truth. Both metrics were computed on binarized vessel maps.

Structure-based Performance. Because curvilinear segmentation emphasizes connectivity and geometric integrity rather than simple region overlap, we also report clDice and Betti error as structure-sensitive measures. The clDice metric evaluates topological alignment between predicted and ground truth skeletons by comparing centerline connectivity (Shit et al., 2021); concretely, it is the harmonic mean of the fraction of the predicted skeleton lying inside the ground-truth mask and the fraction of the ground-truth skeleton lying inside the predicted mask, and thus penalizes broken or disconnected vessels even when overall pixel overlap is high. The Betti error measures the discrepancy in topological invariants specifically, the number of connected components and loops, derived from persistent homology (Hu et al., 2019). A lower Betti error indicates closer structural equivalence between prediction and reference masks. Intuitively, a fragmented vessel inflates the connected-component count, while a broken loop reduces the loop count; Betti error is computed as the absolute difference in these counts between prediction and ground truth, summed over both invariants. Specifically, these topological invariants are extracted using an 8-connectivity convention for foreground structures and a 4-connectivity convention for the background, with image borders zero-padded to prevent artificial disconnections or spurious loops at the boundaries. The final reported Betti error is averaged per image across the respective test dataset. We interpret it jointly with Dice and clDice, since a near-empty prediction can trivially achieve low Betti error while failing on overlap.

Together, these metrics allow us to assess not only how accurately TopoUDA reproduces the target structures in terms of region overlap, but also how well it preserves the underlying structural integrity that defines curvilinear networks.

Threshold-Selection Protocol. All region- and structure-based metrics are computed on binarized predictions. For all methods, the binarization threshold is selected per method on a held-out source-domain validation split, since target labels are unavailable in the UDA setting. The selected threshold is then fixed across all source→target pairs and modalities for that method. For all UDA baselines and TopoUDA, the selected threshold is 0.5 in our experiments. This protocol ensures fair comparison and avoids inadvertently penalizing baselines that produce atypical probability distributions.

4.4. Implementation details

We implemented TopoUDA in PyTorch using the Attention U-Net (Oktay et al., 2018) architecture as the segmentation backbone. The framework was trained on a single NVIDIA A100 GPU with 40 GB memory, requiring approximately 3 h per adaptation task (e.g., STARE→CHASE). The Adam optimizer was employed with an initial learning rate of 1×10^{-4}, reduced by a step decay schedule every 50 epochs. Each experiment was trained for 400 epochs, including a 60 epoch warm-up step using only the labeled source domain before introducing adaptation losses. The balancing weights in the total objective were set as $\lambda_{feat} = 0.1$, $\lambda_{pseudo} = 0.25$, and $\lambda_{adv} = 0.01$ for both discriminators. The persistence thresholds were set as $(\tau_{H_0}, \tau_{H_1}, \tau_{\text{rescue}}) = (0.95, 0.75, 0.65)$. The ProxyNet, a lightweight CNN for differentiable topology estimation, was pre-trained on the source domain for 100 epochs using a mean-squared error objective to approximate the topological feature vector. At inference, TopoUDA processes a single image in 0.5–2 s depending on resolution. For reference, computing ϕ_{topo} via direct persistent-homology evaluation takes approximately 185 ms per image, motivating the use of ProxyNet as a differentiable surrogate during adversarial training.

4.5. Baselines

Source-only trains exclusively on labeled source data and directly infers on the target domain without any adaptation, serving as the lower-bound baseline (reported in Table 2). We compared our method against nine representative unsupervised domain adaptation approaches. FDA (Yang and Soatto, 2020) aligns source and target distributions in the Fourier domain to transfer low-frequency appearance statistics. ADVENT (Vu et al., 2019) utilizes entropy minimization and adversarial learning for output-space alignment. CycleGAN (Zhu et al., 2017) performs unpaired image-to-image translation to render source images in the target appearance style prior to segmentation. CyCADA (Hoffman et al., 2018) performs cycle-consistent image translation combined with feature-level adversarial training. SFDA-DPL (Wu et al., 2023) is a source-free pseudo-label-based method that adapts the model to target data without access to source samples. CADA (Ma et al., 2023) incorporates class-aware adversarial alignment to improve segmentation robustness. CFEA (Wang et al., 2023) introduces feature-enhancement modules tailored for medical segmentation tasks. AdaptDiff (Hul et al., 2024) leverages diffusion-based image translation to synthesize target-like training data conditioned on the source

Table 2
Quantitative comparison of TopoUDA and baseline methods across domain adaptation tasks. Each subtable reports results for a specific modality group. (a) Retinal Fundus experiments include five cross-dataset adaptation pairs. (b) OCTA and EM experiments include four cross-modality and cross-institution pairs. All results are reported as percentages, except Betti error (lower is better). Bold indicates the best performance, underline the second best. TopoUDA consistently achieves the highest Dice, clDice, and lowest Betti error across all scenarios, demonstrating its ability to jointly preserve pixel-level accuracy and topological continuity.

(a) Retinal fundus domain adaptation tasks

Method	STARE → CHASE			DRIVE → CHASE			DRIVE → STARE			CHASE → STARE			CHASE → ROP		
	Dice↑	clDice↑	Betti↓	Dice↑	clDice↑	Betti↓	Dice↑	clDice↑	Betti↓	Dice↑	clDice↑	Betti↓	Dice↑	clDice↑	Betti↓
FDA (Yang and Soatto, 2020)	0.7577	0.4994	0.9777	0.6920	0.4192	1.5004	0.7235	0.4158	1.8422	0.7340	0.4126	1.3289	0.6537	0.3387	0.7482
ADVENT (Vu et al., 2019)	0.7571	0.4942	0.9307	0.6956	0.4180	1.2578	0.7054	0.3776	2.1067	0.7405	0.4399	1.1733	0.6444	0.3263	0.7536
CycleGAN (Zhu et al., 2017)	0.6846	0.3731	0.8373	0.5639	0.1608	1.0364	0.6802	0.3568	2.5756	0.7080	0.3915	1.2067	0.6124	0.2603	0.7618
CyCADA (Hoffman et al., 2018)	0.7462	0.5002	0.8800	0.5408	0.1286	**0.9724**	0.6627	0.3186	2.6911	0.7345	0.4626	1.2244	0.6365	0.3096	0.7353
SFDA-DPL (Wu et al., 2023)	0.7938	0.5336	2.5929	0.7671	0.5219	12.5849	0.8140	0.6282	3.0133	0.7780	0.5502	1.6844	0.6580	0.3378	0.8202
CADA (Ma et al., 2023)	0.7428	0.5614	1.2391	0.6462	0.3686	4.0116	0.6691	0.3661	10.5178	0.5923	0.2100	6.0167	0.6182	0.2633	0.7638
CFEA (Wang et al., 2023)	0.7723	0.5872	**0.6551**	0.6690	0.3929	3.4916	0.7462	0.5665	2.9711	0.7322	0.5453	2.2941	0.5922	0.1808	4.8013
AdaptDiff (Hul et al., 2024)	0.6908	0.4303	6.0667	0.5626	0.2029	37.3667	0.4796	0.1176	3.6611	0.5778	0.5664	3.9500	0.4942	0.1942	10.6417
AIF-SFDA (Li et al., 2025)	0.7838	0.5505	2.5128	0.6498	0.2963	7.5475	0.7149	0.5441	2.2911	0.7237	0.6351	1.0577	0.5331	0.1390	2.3804
TopoUDA (Ours)	**0.8617**	**0.7240**	0.6720	**0.7861**	**0.5906**	1.7600	**0.8216**	**0.7567**	**1.1822**	**0.7804**	**0.6957**	**0.9533**	**0.7356**	**0.4188**	**0.7326**
Source-Only	0.5008	0.4821	3.0453	0.1367	0.1081	1.3200	0.6154	0.6016	1.6600	0.5749	0.5303	1.1778	0.0573	*NaN*	0.8631

(b) OCTA domain adaptation tasks

Method	ROSE → OCTA500			OCTA500 → ROSE		
	Dice↑	clDice↑	Betti↓	Dice↑	clDice↑	Betti↓
FDA (Yang and Soatto, 2020)	0.7495	0.5847	2.5046	0.6587	0.3044	2.6944
ADVENT (Vu et al., 2019)	0.7311	0.5384	3.2407	0.6389	0.2696	2.5138
CycleGAN (Zhu et al., 2017)	0.7401	0.5498	2.7838	0.5720	0.1646	2.9583
CyCADA (Hoffman et al., 2018)	0.7324	0.5024	2.7903	0.6397	0.2794	2.6666
SFDA-DPL (Wu et al., 2023)	0.8147	0.6866	5.3030	0.6659	0.3134	3.76
CADA (Ma et al., 2023)	0.7184	0.4896	3.15	0.7220	0.4498	3.7738
CFEA (Wang et al., 2023)	0.8009	0.6760	3.9903	0.7215	0.4358	2.4236
AdaptDiff (Hul et al., 2024)	0.6542	0.5628	11.9227	0.7111	0.5058	7.3333
AIF-SFDA (Li et al., 2025)	0.8098	0.7393	4.9722	0.7761	0.5369	2.6599
TopoUDA (Ours)	**0.8293**	**0.7853**	**1.8861**	**0.8041**	**0.5403**	**2.3888**
Source-Only	0.5665	0.6130	27.6079	0.4376	0.4376	2.6111

(c) EM domain adaptation tasks

Method	ISBI12 → CREMI			CREMI → ISBI12		
	Dice↑	clDice↑	Betti↓	Dice↑	clDice↑	Betti↓
FDA (Yang and Soatto, 2020)	0.8686	0.9455	0.6949	0.7942	0.8862	1.3671
ADVENT (Vu et al., 2019)	0.8618	0.9447	0.8041	0.7462	0.8678	1.5880
CycleGAN (Zhu et al., 2017)	0.7512	0.9210	1.9022	0.5603	0.8507	2.6109
CyCADA (Hoffman et al., 2018)	0.7476	0.9207	1.9277	0.5179	0.8359	2.6114
SFDA-DPL (Wu et al., 2023)	0.8229	0.9335	0.8904	0.6856	0.8635	1.9885
CADA (Ma et al., 2023)	0.7176	0.8921	2.0793	0.6018	0.8453	2.3270
CFEA (Wang et al., 2023)	0.7574	0.8962	2.9975	0.4909	0.8259	2.5041
AdaptDiff (Hul et al., 2024)	0.6980	0.8302	3.5044	0.7817	0.8141	0.8342
AIF-SFDA (Li et al., 2025)	0.8412	0.9367	1.1781	0.6489	0.8104	2.1755
TopoUDA (Ours)	**0.8879**	**0.9577**	**0.3528**	**0.8158**	**0.8877**	**0.5993**
Source-Only	0.7125	0.8211	0.9370	0.5620	0.7015	2.2786

domain. AIF-SFDA (Li et al., 2025) performs amplitude- and frequency-based feature disentanglement to enable source-free adaptation. For fair comparison, all methods share identical preprocessing and data augmentation. Method-specific hyperparameters (e.g., adversarial loss weights for ADVENT, Fourier mixing ratio for FDA) were tuned following the respective original papers or validated on the source domain to ensure competitive performance.

5. Results

5.1. Quantitative results

Table 2(a–c) summarizes the cross-domain segmentation performance for all baselines and our method across retinal fundus, OCTA, and EM. We additionally report the *Source-Only* baseline, which reflects supervised performance when the model is trained on the source domain and directly evaluated on the target domain. As commonly observed in UDA literature, Source-Only exhibits a pronounced degradation due to severe domain shift, underscoring the difficulty of these transfer scenarios and the necessity of effective adaptation strategies.

Across all nine adaptation tasks, TopoUDA achieves the highest scores in topology-aware metrics (clDice score), while also improving conventional region-based metrics (Dice score). For instance, on CHASE→ STARE, our method improves clDice by almost 0.14 compared with the next-best method, while also achieving 0.7804 Dice. Similar trends are observed in the OCTA-500→ ROSE and CREMI→ ISBI experiments, highlighting that TopoUDA can consistently maintain vessel connectivity and loop correctness across diverse imaging characteristics.

As shown in Table 2, standard natural image UDA baselines (e.g., FDA and ADVENT) partially mitigate appearance differences but struggle to preserve the connectivity of fine vessels. Consequently, their topological scores remain low due to the high incidence of fragmented predictions. Whereas more recent medical UDA approaches (e.g., SFDA-DPL, CFEA, AIF-SFDA) offer improved adaptation, but they remain limited in their ability to capture fine branches and prevent vessel disconnections. We further observe that AIF-SFDA performs competitively on the easier STARE→CHASE task (Dice 0.7838) but degrades sharply on the more challenging CHASE→ROP task (Dice 0.5331, clDice 0.1390), underscoring how recent appearance-focused UDA methods struggle when the source–target gap is dominated by structural rather than purely visual shifts. AdaptDiff, which relies on diffusion-based synthesis, similarly struggles to recover fine branching structure under severe domain shift. TopoUDA surpasses all competing methods in both overlap and topology metrics across modalities without relying on target labels. For the more challenging CHASEDB→ ROP adaptation task, the performance gap between TopoUDA and the baselines is even more pronounced, underscoring the importance of our topological constraints when adapting to a domain with significant structural and quality shifts.

Notably, in a small number of cases, certain baselines appear to match or slightly exceed our betti error values. However, closer inspection reveals that these cases arise when the baseline severely under-segments the vasculature, producing overly sparse predictions that artificially inflate overlap scores despite missing critical vessel connectivity. Betti errors can become deceptively small when a model produces overly sparse or conservative masks, an effect seen in appearance-driven baselines like FDA, CADA, or SFDA-DPL, whose predictions sometimes collapse into a few disconnected blobs. Such trivial topologies naturally produce low Betti-error, even though they miss large portions of the true structure. Our method, in contrast, offers a balanced improvement in both structural completeness and accuracy, and reconstructs a richer, more complete curvilinear network and minor inconsistencies in fine branches or loops can slightly increase the Betti count despite superior anatomical fidelity. This highlights that Betti-error must be interpreted together with overlap metrics and qualitative evidence rather than in isolation.

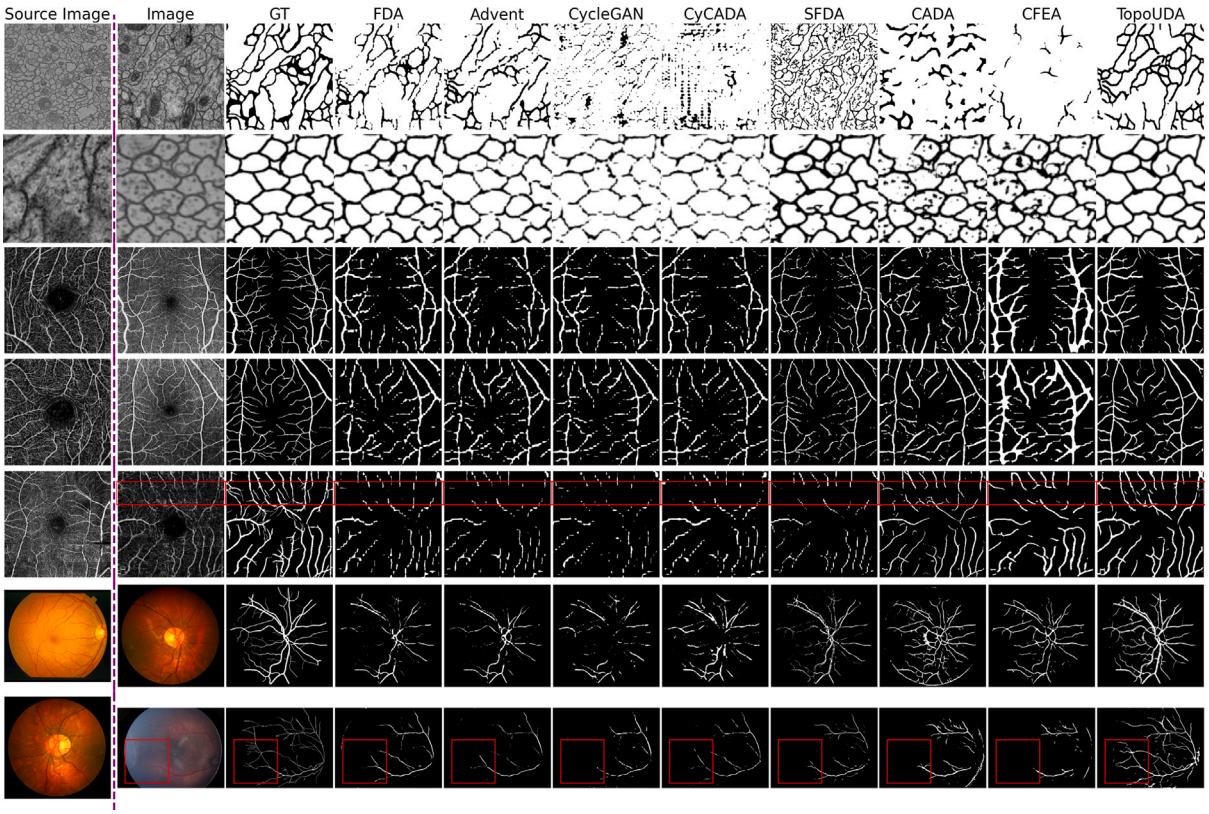


Fig. 5. Qualitative comparison across UDA methods on curvilinear structure segmentation. Each row shows the source image, input target image, and ground truth, followed by outputs from FDA, ADVENT, CycleGAN, CyCADA, SFDA-DPL, CADA, CFEA, and the proposed TopoUDA. Existing UDA methods often produce fragmented, overly sparse, or overly dense predictions depending on the domain e.g., missing elongated structures in OCTA due to local artifacts (red boxes) or generating broken or hallucinatory membranes in EM data. In contrast, TopoUDA consistently reconstructs continuous, confident, and anatomically coherent curvilinear structures, demonstrating its ability to bridge both Appearance and Structural Gaps.

5.2. Qualitative evaluation

Fig. 5 presents representative qualitative comparisons. Baseline methods frequently exhibit broken centerlines, premature branch terminations, and loss of fine vessels, particularly in thin capillary regions and low-contrast areas. These disruptions can severely compromise downstream morphometric analysis and disease assessment.

In contrast, TopoUDA produces globally coherent vasculature, preserving long-range connectivity while recovering thin branches that competing approaches consistently miss. In fundus images, our model reconstructs the peripheral vascular tree and maintains continuity at bifurcations. In OCTA images, TopoUDA avoids gap formation at micro-capillary intersections and delivers more stable connectivity under speckle noise. In EM datasets, our method delineates neuronal membranes without over-connecting background artifacts, highlighting the versatility of our framework across diverse imaging tasks. Overall, TopoUDA reproduces anatomical structures more faithfully across all tested settings, demonstrating clear qualitative advantages over appearance-focused UDA techniques.

5.3. Ablation studies

We perform systematic ablation studies to quantify the contribution of each component in TopoUDA. The ablation studies are conducted on the STARE→ CHASEDB task.

Effect of Different Modules. In Table 3, we provide a component-wise study to evaluate the incremental contribution of each sub-module in TopoUDA, reporting results in terms of Dice, clDice, and Betti error. The term Warm-Up refers to the initial training phase utilizing supervised and feature alignment losses (Section 3.1). D_{pixel} and D_{topo} denote our dual-critic adversarial module for pixel-level and topology-aware discriminators (Section 3.2). HTR denotes the Hierarchical Topological Refiner (Section 3.3), and HTM represents the Hybrid Trust Mechanism (Section 3.4). The results indicate that progressively incorporating each module leads to consistent performance improvements across all evaluation metrics. The Warm-Up baseline achieves the lowest scores, highlighting the difficulty of bridging the structural domain gap when relying solely on supervised and feature-level alignment. Notably, introducing the HTR module results in a simultaneous reduction in Betti error and an increase in clDice, suggesting that the hierarchical refinement strategy effectively enhances vascular connectivity and structural continuity. Adding the dual-critic adversarial modules further improves pixel-level accuracy. While a slight increase in Betti error is observed, likely due to minor topological noise introduced by aggressive texture alignment from D_{pixel}, the inclusion of the topology-aware discriminator D_{topo} mitigates this effect and leads to improved overall structural coherence. Finally, integrating the Hybrid Trust Mechanism (HTM) yields the best overall performance, validating the effectiveness of trust-weighted supervision in stabilizing pseudo-label learning.

Last two rows before the complete our run disentangle the role of each discriminator when HTR and HTM are present. Removing D_{pixel} leaves Dice nearly unchanged (−0.002) but raises Betti error by 0.059 over the full model, while removing D_{topo} degrades clDice by 0.025 and Betti by 0.078. D_{topo} is therefore the dominant driver of topological fidelity, while D_{pixel} contributes complementary appearance alignment. The dip in 6th row (without HTM) is fully recovered once HTM is reinstated, confirming that HTM is the integrator that reconciles HTR-derived pseudo-labels with the discriminators.

Hyperparameter Sensitivity. To assess whether TopoUDA's performance depends critically on the choice of loss-weighting coefficients,

Table 3
Ablation study on the contribution of each module in TopoUDA. D_{pixel}: Pixel-Level Discriminator, D_{topo}: Curvilinear-Aware Discriminator, HTR: Hierarchical Topological Refiner, CAD: Curvilinear-Aware Discriminator, HTM: Hybrid Trust Mechanism. Bold indicates the best performance.

Warm-Up	D_{pixel}	D_{topo}	HTR	HTM	Dice↑	clDice↑	Betti↓
✓					0.7877	0.6360	1.133
✓	✓				0.8247	0.6409	1.585
✓		✓			0.8258	0.6728	0.8499
✓	✓	✓			0.8503	0.7064	0.8374
✓			✓		0.8542	0.7067	0.7865
✓	✓	✓	✓		0.8400	0.6765	0.7760
✓		✓	✓	✓	0.8598	0.7172	0.7305
✓	✓		✓	✓	0.8503	0.6990	0.7500
✓	✓	✓	✓	✓	**0.8617**	**0.7240**	**0.6720**

Table 4
Hyperparameter, loss-weight, sensitivity analysis on STARE→CHASE. One weight is varied while the others are held at the default ($\lambda_{\text{feat}} = 0.1$, $\lambda_{\text{adv}} = 0.01$, $\lambda_{\text{pseudo}} = 0.25$, shown in bold). TopoUDA is robust to ±50% perturbations.

Variant	Setting	Dice↑	clDice↑	Betti↓
λ_{feat}	0.05	0.8635	0.7152	0.7050
	0.10 (default)	**0.8617**	**0.7240**	**0.6720**
	0.20	0.8417	0.6860	0.9520
λ_{adv}	0.005	0.8509	0.7259	0.9110
	0.01 (default)	**0.8617**	**0.7240**	**0.6720**
	0.020	0.8458	0.6933	0.9635
λ_{pseudo}	0.10	0.8588	0.7118	0.6965
	0.25 (default)	**0.8617**	**0.7240**	**0.6720**
	0.50	0.8499	0.7161	0.6970

we sweep each of λ_{feat}, λ_{adv}, and λ_{pseudo} around its default value while holding the others fixed (Table 4). Across all settings, Dice varies within roughly two percentage points of the default and clDice within four, indicating that the framework is robust to moderate perturbations of the weighting scheme. The default configuration ($\lambda_{feat} = 0.10$, $\lambda_{adv} = 0.01$, $\lambda_{pseudo} = 0.25$) yields the best overall balance across the three metrics; while an individual setting occasionally edges out the default on a single metric (e.g. $\lambda_{feat} = 0.05$ on Dice), no configuration improves all three jointly, supporting the choice of these values throughout the main experiments.

Threshold Sensitivity. To verify that the choice of binarization threshold (0.5) does not advantage any particular method, we sweep the threshold over $[0.3, 0.7]$ and recompute all three metrics for TopoUDA and the two strongest baselines on STARE → CHASE (Table 5). TopoUDA is largely insensitive to threshold choice, with Dice varying by 0.20 points (0.8599–0.8619) and clDice by 2.3 points (0.7106–0.7337) across the range. SFDA-DPL and CFEA show larger swings (1.6 and 3.1 Dice points respectively) with a clear monotonic decline as the threshold rises, indicating that their probability distributions favor a threshold below 0.5. Even at each baseline's best per-metric threshold, TopoUDA still outperforms both on Dice and clDice across the full sweep.

Comparison with Topology-Aware Source-Only Training. A natural question is whether simple topology-aware losses, applied at supervised training time, can close the structural domain gap on their own, without an adaptation framework. We test this by augmenting the Source-Only baseline with two representative differentiable topology losses: clDice (Shit et al., 2021) and the persistent-homology-based TopoLoss of (Hu et al., 2019). Both variants are trained on labeled source data only and directly evaluated on the target domain. Results on the two most distinct fundus tasks are reported in Table 6. Topology losses substantially improve Dice over the vanilla Source-Only baseline (+0.16 on STARE → CHASE; +0.54 on CHASE → ROP), confirming that they recover coarse vessel structure when no target-domain signal is available. However, the improvement does not consistently extend to topological metrics: clDice and Betti error often worsen on the target domain, reflecting that source-trained skeleton or persistence priors do not transfer cleanly under appearance shift. TopoUDA outperforms both topology-loss variants by a large margin on all metrics, indicating that explicit modeling of the source–target gap, rather than topology supervision alone, is what enables consistent structural alignment.

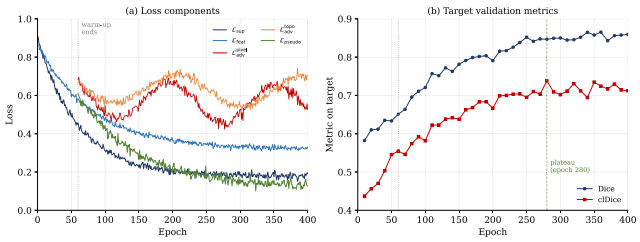


Fig. 6. Evolution of the individual loss terms over 400 training epochs on STARE → CHASE. Adaptation losses are introduced gradually after the warm-up phase (vertical dashed line at epoch 60). The two discriminator losses (bottom row) remain near $\log 2$ throughout adversarial training, confirming a stable equilibrium between the generator and both critics.

Training Dynamics. Adversarial training in unsupervised domain adaptation is known to be susceptible to instabilities such as mode collapse and discriminator dominance; we mitigate these risks through a supervised warm-up, a small adversarial weight ($\lambda_{\text{adv}} = 0.01$), and dual-critic separation (D_{pixel} and D_{topo} operate on disjoint feature spaces). To provide insight into the optimization behavior of the full objective, Fig. 6 visualizes the evolution of each loss term over the 400-epoch training schedule on STARE → CHASE. The supervised segmentation loss decreases rapidly during the warm-up phase (epochs 0–60), after which the feature-alignment, pseudo-label, and dual-critic adversarial losses are introduced. Both the pixel- and topology-discriminator losses stabilize close to $\log 2 \approx 0.69$ for the remainder of training, indicating that neither critic dominates the generator and that the adversarial game converges to a stable equilibrium rather than collapsing to a trivial solution. Target-domain metrics plateau by approximately epoch 280; we nonetheless train to the full 400-epoch budget to ensure the adversarial equilibrium is stably maintained, at negligible additional cost (~3 h total per task).

Impact of Pseudo-Labeling Strategy (HTR). We further evaluate the effectiveness of our pseudo-labeling design by comparing it with a conventional confidence-based approach. We compare traditional confidence-based pseudo-labeling using various thresholds against our HTR method. To ensure a fair comparison, we fix all other modules (CAD, HTM) and replace only the HTR with the confidence-based baseline. As shown in Table 7, the traditional scheme reveals a critical limitation: the optimal threshold varies across metrics, forcing a trade-off between structural completeness and topological correctness. A low threshold (0.3) achieves the highest baseline clDice (0.6641) by capturing faint vessels, but at the cost of significant topological noise (Betti error 2.80). Conversely, a high threshold (0.9) minimizes false loops (Betti error 1.45) but fragments the network, degrading clDice (0.6176). A balanced threshold (0.5) maximizes overlap (Dice) but fails to optimize either structural metric. This demonstrates that pixel-level confidence cannot simultaneously satisfy connectivity and topology constraints. In contrast, our HTR achieves the best performance across all three metrics, demonstrating that it successfully breaks the trade-off between connectivity and topological noise.

To illustrate the qualitative impact of our refinement strategy, Fig. 7 compares traditional confidence-based pseudo-labels with our HTR-generated pseudo-labels on representative fundus (CHASEDB) and EM (ISBI12) examples. In both modalities, raw probability maps contain low-confidence regions at structurally critical locations, such as vessel junctions and membrane intersections. Traditional thresholding fails

Table 5
Threshold sensitivity on STARE → CHASE. The binarization threshold is varied from 0.3 to 0.7; the default value used in all main experiments is 0.5. Best value per method per metric is shown in bold.

Thr.	TopoUDA (Ours)			SFDA-DPL (Wu et al., 2023)			CFEA (Wang et al., 2023)		
	Dice↑	clDice↑	Betti↓	Dice↑	clDice↑	Betti↓	Dice↑	clDice↑	Betti↓
0.3	0.8615	**0.7337**	0.7065	**0.8016**	**0.5502**	2.5964	**0.7855**	**0.6138**	0.6773
0.4	**0.8619**	0.7287	0.6915	0.7976	0.5418	2.5928	0.7791	0.6007	0.6915
0.5	0.8617	0.7240	0.6720	0.7938	0.5336	2.5929	0.7723	0.5872	0.6551
0.6	0.8612	0.7182	0.6570	0.7898	0.5253	**2.5795**	0.7645	0.5715	**0.6284**
0.7	0.8599	0.7106	**0.6510**	0.7852	0.5166	2.6071	0.7543	0.5501	0.6453

Table 6
Comparison against Source-Only training augmented with topology-aware losses on two fundus adaptation tasks. Topology losses recover coarse structure (Dice) but do not consistently transfer connectivity (clDice) or persistent topology (Betti) across the domain gap.

Method	STARE → CHASE			CHASE → ROP		
	Dice↑	clDice↑	Betti↓	Dice↑	clDice↑	Betti↓
Source-Only	0.5008	0.4821	3.0453	0.0573	*NaN*	0.8631
Source-Only + clDice (Shit et al., 2021)	0.6571	0.3545	1.6097	0.6017	0.1829	4.6200
Source-Only + TopoLoss (Hu et al., 2019)	0.6469	0.3482	1.4195	0.5873	0.1557	4.5331
TopoUDA (Ours)	**0.8617**	**0.7240**	**0.6720**	**0.7356**	**0.4188**	**0.7326**

Table 7
Ablation study on traditional pseudo-labeling with varying confidence thresholds compare to our strategy.

Threshold	Dice↑	clDice↑	Betti↓
0.3	0.8154	0.6641	2.8071
0.5	0.8201	0.6359	1.9431
0.9	0.8136	0.6176	1.4560
Our HTR	**0.8617**	**0.7240**	**0.6720**

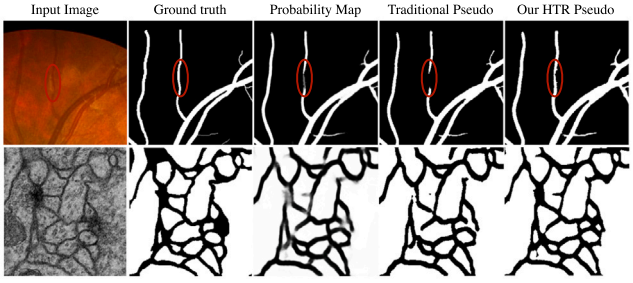


Fig. 7. Effect of Hierarchical Topological Refinement (HTR) on pseudo-label quality. Examples from CHASEDB (top) and ISBI12 (bottom) showing input, ground truth, probability map, pseudo-labels from confidence thresholding, and pseudo-labels from HTR. While traditional pseudo-labels drop low-confidence but structurally important regions, HTR restores these segments via persistence-based anchor detection and spatial refinement, producing globally coherent pseudo-labels.

Table 8
Ablation study on our proposed HTR with varying topological thresholds.

H_0 Threshold	H_1 Threshold	Dice↑	clDice↑	Betti↓
0.5	0.5	0.8224	0.6594	0.7990
0.95	0.5	0.8274	0.6629	1.8897
0.5	0.75	0.8259	0.6490	1.4559
0.95	0.75	**0.8617**	**0.7240**	**0.6720**

to recover these low-confidence structurally critical regions, producing fragmented masks. In contrast, HTR identifies these topologically important structures and successfully restores their continuity. This highlights HTR's ability to preserve global connectivity even when the underlying probability map is locally ambiguous.

Next, we analyze the sensitivity of our proposed HTR to its own topological thresholds in Table 8. Our method demonstrates robust performance across different settings. The optimal performance is achieved with a high threshold for connected components ($H_0 = 0.95$) to prune noise, and a moderate threshold for loops ($H_1 = 0.75$) to preserve anatomical structures, validating our methodological design choices.

Table 9
Ablation of homology components in pseudo-label refinement.

Threshold Style	Dice↑	clDice↑	Betti↓
H_0-only	0.8459	0.6898	0.8705
H_1-only	0.8570	0.7182	0.7055
Ours	**0.8617**	**0.7240**	**0.6720**

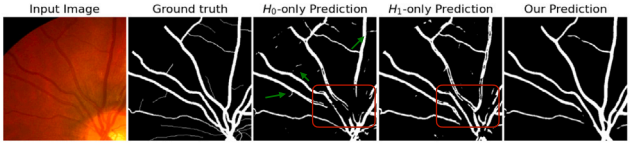


Fig. 8. Ablation of homology components in pseudo-label refinement. H_1-only preserves long continuous structures (red box) but misses fine branches while H_0-only captures small fragments (green arrows) but produces globally incomplete branches.

Lastly, to further dissect the role of topological cues in pseudo-label construction, we conduct an additional ablation isolating the two homology channels of our refiner: H_0-only pseudo-labels, which preserve connected-component–based branch fragments, and H_1-only pseudo-label, which retain loop-based high-persistence anchors. This analysis evaluates whether each topological cue alone is sufficient for structurally coherent supervision, and whether their combination in our full HTR design is indeed necessary. We utilize the optimal thresholds identified above ($H_0 = 0.95, H_1 = 0.75$) for this comparison. As shown in Table 9, the H_1-only variant achieves higher Dice and clDice than the H_0-only counterpart (0.8570 vs. 0.8459 Dice; 0.7182 vs. 0.6898 clDice), reflecting its advantage in preserving long continuous trajectories, whereas H_0-only attains a slightly lower Betti error by suppressing spurious loops. Fig. 8 follows the same pattern: H_0-only pseudo-labels tend to produce short, disconnected segments, while H_1-only pseudo-labels exhibit strong global continuity but neglect fine-scale structure. These findings confirm that combining H_0 and H_1 is useful for producing pseudo-labels that are both topologically stable and anatomically complete.

Impact of the Hybrid Trust Mechanism (HTM). To evaluate the reliability of HTM, we analyze the relationship between hybrid trust

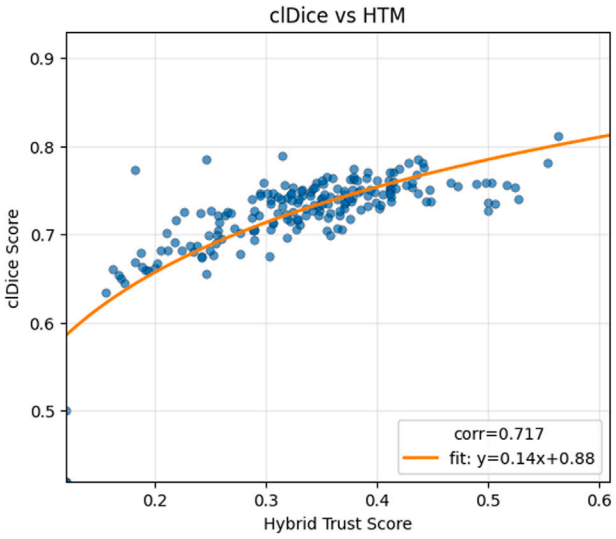


Fig. 9. Scatter plot of hybrid trust scores versus clDice across the target domain. A strong positive relationship (Pearson r = 0.717) demonstrates that HTM reliably ranks pseudo-labels by structural accuracy. The saturating curve indicates effective suppression of low-quality pseudo-labels and consistent performance for high-trust predictions.

Table 10
Backbone analysis of TopoUDA. Comparison of segmentation backbones on cross-domain vessel adaptation on task of STARE → CHASE, tested on CHASE test set. Our topology-aware modules generalize consistently across architectures.

Backbone	Dice↑	clDice↑	Betti↓
U-Net (Ronneberger et al., 2015)	0.8356	0.6937	1.8062
Attention U-Net (Oktay et al., 2018)	**0.8617**	**0.7240**	0.6720
ResU-Net (Diakogiannis et al., 2020)	0.8572	0.6998	**0.6575**

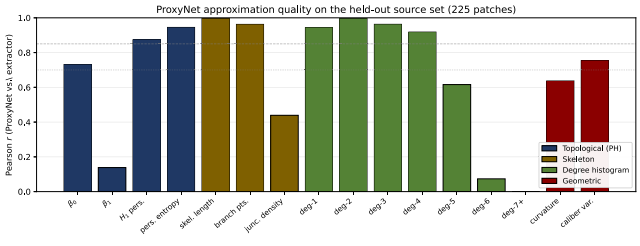


Fig. 10. ProxyNet approximation quality on the held-out CHASE source set. Per-feature Pearson correlation r between ProxyNet predictions and ground-truth $\tilde{\phi}_{\text{topo}}$ values, colored by feature group. The dashed and dotted reference lines mark $r = 0.85$ and $r = 0.70$, respectively. The deg-7+ bar is shown as a hatched placeholder, since this bin is constant zero across the test set (correlation undefined).

scores and structural accuracy on the target domain. Fig. 9 visualizes a scatter plot of hybrid trust scores versus the corresponding clDice values. We observe a positive association between trust and structural accuracy (Pearson r = 0.717), indicating that HTM produces a quality signal that is highly predictive of topological correctness. Notably, the trend exhibits a steep rise at low trust scores followed by a plateau, reflecting a desirable behavior: HTM effectively suppresses structurally unreliable pseudo-labels (sharp performance gain when low-trust samples are avoided), while high-trust samples are uniformly accurate, resulting in diminishing returns at the upper end. This pattern confirms that HTM successfully ranks pseudo-labels by anatomical plausibility and that the combined pixel- and topology-based scores provide a more discriminative reliability measure than either cue alone.

Backbone Robustness. We evaluate TopoUDA across three segmentation backbones (U-Net, Attention U-Net, ResUNet) on the STARE → CHASE task to validate its generality (Table 10). Across all three backbones, TopoUDA demonstrates consistent performance, achieving an average Dice of 0.8515 and clDice of 0.7058. Interestingly, the U-Net backbone exhibits slightly higher Betti error despite comparable segmentation accuracy. This discrepancy reflects the model's enhanced sensitivity: whereas conservative predictions often miss fine vessels entirely, yielding sparse outputs with deceptively low topological complexity, U-Net successfully captures intricate vascular networks. This increased capture rate inevitably introduces minor topological noise (e.g., small loops) inherent to high-fidelity extraction, reflecting structural completeness rather than failure. These results confirm that TopoUDA functions as a robust, backbone-agnostic adaptation framework. We also note that baseline methods in our main comparison employ their own specialized architectures rather than vanilla U-Net; thus, architectural differences may partially account for performance variations observed in Table 2.

Proxy Network Approximation Quality. Because ProxyNet provides the differentiable bridge between the segmentation network and the topology discriminator D_{topo}, we analyze how faithfully it approximates the ground-truth fingerprint ϕ_{topo} on a held-out source set (CHASE, 225 patches). For each of the 16 descriptor dimensions, we report the Pearson correlation r between ProxyNet predictions $P_\psi(y)$ and the standardized ground-truth values $\tilde{\phi}_{\text{topo}}(y)$, along with the mean squared error. Per-group results are summarized in Table 11, and the per-feature breakdown is shown in Fig. 10. Eleven of the sixteen features achieve a positive correlation of $r \geq 0.60$, demonstrating broad signal coverage across all four feature groups: persistent-homology descriptors (total H_1 persistence $r = 0.88$, persistence entropy $r = 0.95$, β_0 $r = 0.73$), skeleton-geometry features (skeleton length $r = 1.00$, branch-point count $r = 0.96$, degree-2/3/4 counts $r = 1.00/0.96/0.92$, degree-1 $r = 0.94$), and geometric descriptors (caliber variation $r = 0.74$, mean curvature $r = 0.64$). The four lower-correlation features (β_1, deg-5, deg-6, junction density) are not failure cases of the surrogate but reflect properties of the specific source domain used here (CHASE retinal fundus): β_1 is sparse because adult retinal vasculature on 400 × 400 crops rarely contains closed arcades, and high-valence degree bins are rarely populated in fundus skeletons. Their per-feature predictiveness is expected to be substantially higher on modalities where these structures are prevalent, for example, β_1 in EM neuron membranes (mean β_1 of 282/662 per image; see Table 1) or in OCTA capillary networks. The per-feature quality reported here is therefore a domain-specific snapshot rather than an intrinsic limit on ProxyNet's representational capacity. ProxyNet is used as a differentiable gradient source rather than as an oracle estimator of ϕ_{topo}, and the strong downstream performance reported throughout this paper across all nine adaptation tasks confirms that the gradients produced by ProxyNet are sufficient to drive structural alignment across these diverse modalities.

Comparison with a Pretrained Perceptual Proxy. To examine whether a task-specific lightweight proxy is necessary, or whether off-the-shelf pretrained features suffice, we replace ProxyNet with a proxy built on a frozen ImageNet-pretrained VGG-16 backbone followed by a small linear regression head, in the spirit of early supervised topology-aware work (Mosinska et al., 2018). The two proxies are pre-trained under identical conditions on source ground-truth masks and plugged into the full TopoUDA framework on STARE → CHASE. As shown in Table 12, the two proxies have comparable trainable cost: ProxyNet trains 106,574 parameters end-to-end, while the VGG-16 variant trains only the 33,872 parameters of the linear regression head on top of a 14.7 M frozen backbone. Despite this similar trainable footprint, ProxyNet outperforms the VGG-16 variant on all three downstream metrics, with the largest gap on Betti error (a reduction of approximately 41%). The VGG-16 features encode generic perceptual statistics tuned for natural-image classification and do not directly represent

Table 11
ProxyNet per-feature approximation quality on the held-out CHASE source set (descriptors standardized to zero mean and unit variance). MSE values close to zero indicate accurate approximation.

Feature group	Mean Pearson r	Mean MSE	Worst-case feature
Topological (4 features)	0.686	0.395	β_1 ($r = 0.19$)
Skeleton (3 features)	0.788	0.309	junc. density ($r = 0.40$)
Degree histogram (7 features)	0.771	0.251	deg-7+ (r undef.)
Geometric (2 features)	0.688	0.538	curvature ($r = 0.64$)
Overall (16 features)	**0.740**	**0.334**	–

Table 12
Comparison of differentiable topology proxies under the full TopoUDA framework on STARE → CHASE. "# Params" refers to the proxy module only and reports both total and trainable parameter counts. The VGG-16 backbone is frozen at ImageNet-pretrained weights; only the linear head is trained. **Bold** = best.

Proxy variant	Total params	Trainable params	Dice↑	clDice↑	Betti↓
VGG-16 + Linear Head	14.7 M	33,872	0.8438	0.7089	1.1310
ProxyNet (ours)	**0.107 M**	**106,574**	**0.8617**	**0.7240**	**0.6720**

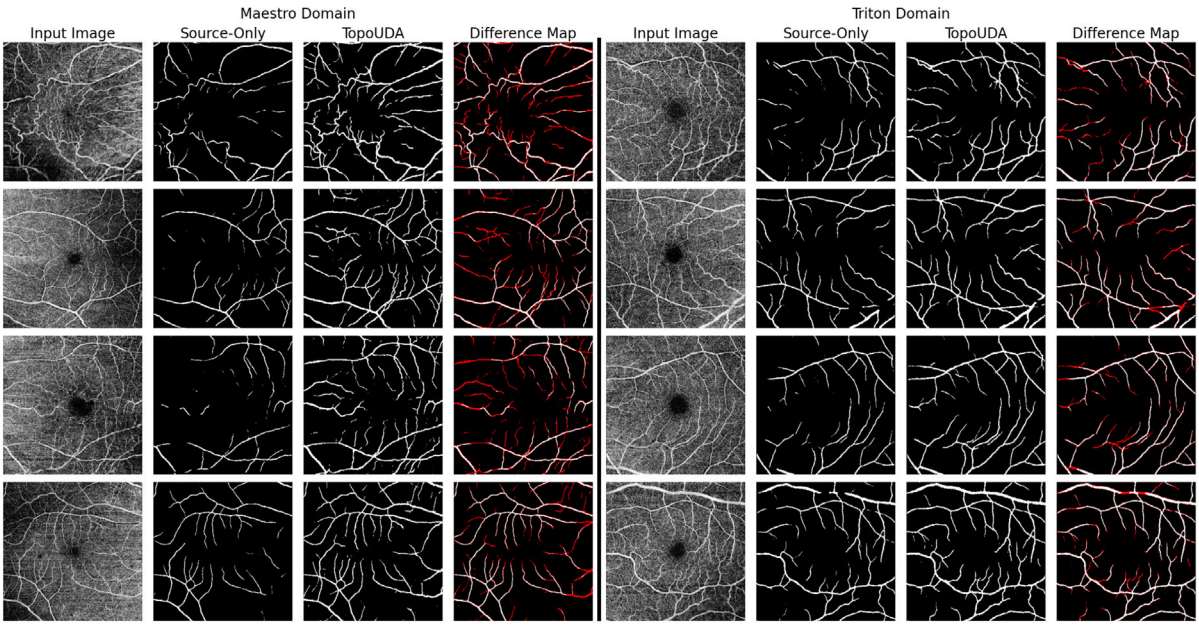


Fig. 11. Qualitative generalization on the unlabeled AI-READI dataset. The model is adapted from the OCTA-500 source domain to two distinct target device domains: Topcon Maestro (Columns 1–4) and Triton (Columns 5–8). Columns display for each domain: Input OCTA projection; Source-Only baseline (U-Net trained on source); TopoUDA (Ours); and Difference Map, where vascular structures recovered by TopoUDA but missed by the baseline are highlighted in red. Note that while the baseline suffers from significant fragmentation and loss of fine capillaries due to domain shift, TopoUDA successfully restores global vessel connectivity and preserves intricate branching structures across both scanner types.

the explicit topological quantities composing ϕ_{topo} (Betti numbers, persistence entropy, degree-distribution counts), whereas ProxyNet is supervised directly on these targets. This supports the choice of a small task-specific proxy over a generic pretrained perceptual encoder: at comparable trainable parameter cost, supervising directly on the structural targets yields a more accurate and topologically reliable gradient source.

5.4. Qualitative generalization AI-Readi

To assess the practical utility of TopoUDA in real-world deployment scenarios, we applied the framework to the AI-READI dataset (AI-READI Consortium, 2025), which lacks pixel-level annotations. We established two distinct UDA settings adapting from the OCTA-500 source domain to two specific device domains within AI-READI: Topcon Maestro and Triton. Fig. 11 presents a visual comparison between a standard "Source-Only" baseline (Attention U-Net trained on OCTA-500) and TopoUDA results, where the first four columns correspond to the Maestro domain and the subsequent four columns to the Triton domain.

We observe that the Source-Only baseline exhibits significant structural degradation, characterized by fragmented main vessels and a failure to capture fine capillary branches. In contrast, TopoUDA demonstrates superior structural sensitivity, successfully recovering intricate small branches while maintaining the continuous connectivity of the main vascular tree. This improvement is explicitly visualized in the difference maps (Fig. 11, last column), where vascular structures recovered by TopoUDA but missed by the baseline are highlighted in red. These highlights confirm that the adaptation module actively bridges connectivity gaps and restores fine-grained details, supporting the framework's feasibility for large-scale, heterogeneous clinical studies. They also offer a practical contribution: the high-quality segmentations produced without any target-domain annotations can serve as initial labels for human-in-the-loop (HIL) refinement, reducing

manual annotation burden and accelerating ground-truth creation for downstream OCTA biomarker studies in type 2 diabetes.

5.5. Failure analysis

While TopoUDA consistently improves structural preservation across all datasets, certain challenging cases remain. Fig. 12 illustrates representative scenarios in which extreme imaging conditions or annotation-specific factors lead to localized errors. These failure modes highlight the limitations of structure-aware adaptation when curvilinear cues are weak, ambiguous, or absent. These cases primarily arise in:

(i) ROP fundus images, where immature, irregular neovascular branches that are only faintly visible even in the raw image, leading to local disconnects. In such regions, TopoUDA correctly reconstructs the main vasculature but may miss subtle terminal branches, producing small false negatives. These errors arise when the local contrast falls below the persistence thresholds used for anchor extraction and branch refinement. Importantly, the missed structures are visually ambiguous even for human annotators and do not compromise the global vessel tree.

(ii) Deep capillary plexus OCTA layers, where flow intensity fluctuations are caused by flow dropout, projection artifacts, or motion noise. These errors are spatially localized and typically occur in peripheral or low-perfusion regions

(iii) EM boundary maps, where ground-truth annotations sometimes include interior fill regions between membrane boundaries. Because these filled zones do not manifest curvilinear structure, they have negligible topological persistence and minimal skeleton support. TopoUDA therefore treats them as structurally insignificant and often leaves them unsegmented, resulting in conservative under-segmentation inside closed boundaries. Notably, the membrane contours themselves remain continuous and accurate; the failure is specific to annotation conventions rather than structural misinterpretation. This modality-specific behavior reflects TopoUDA's bias toward filamentary geometry, suggesting that future extensions may integrate multi-region or multi-topology cues for improved compatibility with diverse annotation schemes.

Across all three modalities, TopoUDA's errors remain localized rather than systemic, and the global organization of the curvilinear network is preserved. Despite these limitations, TopoUDA consistently outperforms baselines, which typically exhibit aggressive under-segmentation with severe branch loss under the same conditions. Future work may improve robustness by incorporating uncertainty modeling, adaptive persistence thresholds, or multi-scale structural priors.

6. Discussion and conclusion

In this work, we presented TopoUDA, a unified structure-aware unsupervised domain adaptation framework for curvilinear structure segmentation across diverse medical imaging modalities. By integrating a dual-critic adversarial system, a hierarchical topology-guided pseudo-label refiner, and a hybrid trust mechanism, our approach jointly enforces visual realism and structural coherence during adaptation. Extensive experiments on retinal fundus, OCTA, and EM datasets demonstrate that TopoUDA consistently improves cross-domain generalization and achieves state-of-the-art performance on both overlap-based and topology-aware metrics. These results highlight the effectiveness of incorporating topological and geometric priors into the UDA pipeline and underscore the framework's ability to produce structurally faithful segmentations in label-scarce clinical environments.

Despite these strong improvements over existing UDA methods, we acknowledge some limitations that warrant further investigation. The topology-guided refinement depends on the stability of persistent features, and extremely low-contrast or physically disrupted regions may yield pseudo-labels that are conservative in restoring very fine structures. Similarly, the ProxyNet-based differentiable topology approximation, although efficient and effective in practice, abstracts away some fine-grained geometric nuances. Importantly, these behaviors primarily occur in inherently ambiguous regions and do not undermine the overall structural reliability achieved by our method.

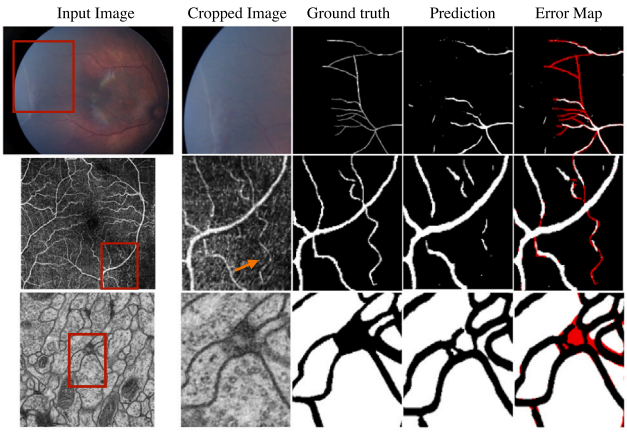


Fig. 12. Representative failure cases of TopoUDA across three modalities: Retinal Fundus (top row), OCTA (middle row), and EM (bottom row). The red contours in the Error Map highlight missing segmentations (false negatives), which typically occur in regions with extremely low contrast or ambiguous structural boundaries.

Looking ahead, the insights gained from these observations suggest several natural extensions. Incorporating complementary cues such as multi-scale context or uncertainty estimates may help further stabilize refinement in challenging regions. Additionally, exploring lightweight alternatives to ProxyNet or broader families of topology-aware descriptors may provide even greater flexibility across modalities. While these directions remain outside the scope of this work, they represent promising avenues for building on the strengths of TopoUDA.

Overall, TopoUDA represents a significant step toward structure-aware domain adaptation for curvilinear segmentation. By explicitly modeling both appearance and structural cues, it reduces reliance on target-domain annotations and offers a practical, robust solution for deploying segmentation models across heterogeneous clinical settings.

CRediT authorship contribution statement

Gözde Merve Demirci: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Chia-Ling Tsai:** Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization. **Jiaqi Yang:** Writing – review & editing, Writing – original draft, Conceptualization. **Chao Chen:** Formal analysis, Conceptualization. **Wei-Chi Wu:** Data curation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Gemini LLM in order to improve readability and check grammar. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The public and AI-READI datasets are accessible via cited links. In-house clinical data is confidential due to privacy. The TopoUDA source code will be made public on GitHub upon acceptance.

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